

Milestone 1 - Project Proposal

Group 6

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Application/Website Idea

The idea of our project is to create a top music chart website that showcases the top 50 songs from different countries and genres around the world. The website will let users compare international charts, explore artist and genre insights, and visualize music trends over time. The website will also feature personalized recommendations of genres and artists based on information that a user fills out in their profile.

Datasets

1. Dataset 1 - Worldwide Music Artists Dataset

Description: This dataset features detailed information on over 100,000 artists across various genres and countries, including their names, associated genres, Spotify IDs, locations, and profile images. It's an ideal resource for analyzing global music trends, artist demographics, and genre distribution.

Link:

<https://www.kaggle.com/datasets/harshdprajapati/worldwide-music-artists-dataset-with-image?select=Global+Music+Artists.csv>

Relevant Size Statistics:

- **Number of rows:** 101,375
- **Number of attributes:** 4
- **Memory usage:** 37.09 MB

Summary Statistics of Several Attributes:

- **Number of Artists:** 101,375

- **Number of Unique Countries:** 188
- **Number of Artist Genres:** 27,959
- **Most represented country:** United States (33,032 artists)
- **Most represented genre:** hip hop (35,404 artists)

Top 5 Countries by Number of Artists:

Country	Number of Artists
United States	33,032
United Kingdom	12,829
Germany	6,703
France	4,279
Canada	3,447

Top 5 Genres:

Country	Number of Artists
hip hop	35,404
movie tunes	137
video game music	72
jazz saxophone	71
turkish pop	69

2. Dataset 2 - Spotify Top 50 Playlist Songs

Description: This dataset contains daily snapshots of the Top 50 most popular songs on Spotify across multiple countries (including the US, UK, Japan, and others), featuring detailed metadata such as song title, artist(s), popularity score, release date, duration, and explicit content flags. The data is collected automatically each day via a scheduled GitHub workflow using Spotify's API.

Link: <https://www.kaggle.com/datasets/anxods/spotify-top-50-playlist-songs-anxods/data>

Relevant Size Statistics:

- **Number of rows:** 27,800 x 9 countries = 250,200
- **Number of attributes:** 11
- **Memory usage:** 12.08 * 9 countries = 108.72 MB

Summary Statistics of Several Attributes:

- **Number of Songs:** 27,800
- **Number of Unique Songs:** 391
- **Number of Unique Artists:** 241
- **Date Range:** 2023-05-18 to 2024-11-27
- **What percentage of songs are explicit?** 8889 / 27800 (32%)

As seen below, the mean position of a song is 25.5 with a standard deviation of 14.4. In addition, the mean popularity of a song is 79.5 with a standard deviation of 9.3. The average duration of a song is approximately 3 minutes long (181,741ms) with a standard deviation of 47 seconds (47,587ms).

	Position	Popularity	Duration_ms	Total_tracks
Count	27,800	27,800	27,800	27,800
Mean	25.5	79.5	181,741	5
SD	14.4	9.3	47,587	6
Min	1	0	68,874	1
25%	13	74	153,609	1
50%	25	79	171,667	1
75%	38	85	198,495	10
max	50	100	500,989	40

Queries

1. Compare the performance of artists' songs in their home country vs internationally
2. Examine the performance of songs in each country that are from domestic vs international artists
3. For each country, find the most common genres of artists appearing in the Top 50 playlists, and the average popularity of songs in those genres.
4. For a selected group of genres, analyze how the average song popularity has changed over time in each country.
5. Select album_type if it is a single or album. If it is an album then return the number of total_track in the album.
6. For each top 50 songs, return the number of songs that is_explicit, to see if the most popular songs are explicit or not.
7. Compare the Top 50 Songs of one country to another country/worldwide.
8. For a selected genre/group of genres and country look at the average time of a song.